DESIGNING A COST-EFFECTIVE LINE HAUL ROUTE NETWORK WITH HETEROGENEOUS VEHICLES

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- *What is line haul route network design?*

- *Why is it important to consider heterogeneous vehicles when designing line haul routes?*

Line Houle Route Network Design

Line haul route network design is the process of planning and optimizing the movement of goods between different locations. involves determining the best routes for vehicles to travel, the types of vehicles to use, and the frequency of deliveries.

Heterogenious Vechicles

Heterogeneous vehicles are vehicles of different types, different capacities, speeds, and ranges.

Why is it important to consider heterogeneous vehicles when designing line haul routes?

- o reduce costs: Heterogeneous vehicles can be used to reduce the overall cost of line haul transportation. For example, smaller vehicles can be used for shorter routes or for routes with lower demand, while larger vehicles can be used for longer routes or for routes with higher demand.
- To improve fuel efficiency: Heterogeneous vehicles can be used to improve the overall fuel efficiency of line haul transportation. For example, smaller vehicles are typically more fuel-efficient than larger vehicles.
- To reduce emissions: Heterogeneous vehicles can be used to reduce the overall emissions of line haul transportation. For example, electric vehicles can be used to reduce emissions from tailpipes.
- To improve flexibility: Heterogeneous vehicles can be used to improve the overall flexibility of line haul transportation. For example, if a vehicle breaks down, another vehicle can be used to take its place.

Parameters

> Origin Destination Pairs .

>Time Frame (15 Days).

>Heterogeneous fleet of vehicles.

>Due time (time in hours available from pickup at origin hub till its destination).

> OD pair load (load volume between OD pairs for 15 days).

>Cost of ad hoc routes (assumed to be 30% higher than that of the regular routes).

Assumptions

- The handling time at cross-docks and stopovers is the same for all vehicle types.
- There is no traffic congestion.
- The vehicles are always available.
- The average speed of trucks on the network is 50 km/hour.
- The time required for processing cross docking shipments and stopovers at each facility is 30 minutes.
- The heterogeneous fleet of vehicles consists of 4 vehicle types.

Methodology

We have adopted Mixed Integer Programming for solving the problem of designing set of routes and ad hoc heterogenious vechicle routes to meet various origin- destination demand requirements.

Code

```
import numpy as np
import pandas as pd
from ortools.linear_solver import pywraplp
# Load the data set
df = pd.read_excel("https://1drv.ms/x/s!AlXbkEwavu3f0WZfEPe_ypqcbiq4?e=YmEycA")
# Define the problem parameters
num_vehicle_types = 4
num_days = 15
num_hub_cities = 50
# Create the solver
solver = pywraplp.Solver('line_haul_routing', pywraplp.Solver.CBC_MIXED_INTEGER_PROGRAMMING)
# Define the decision variables
x = np.zeros((num_vehicle_types, num_hub_cities, num_days), dtype=np.int8)
y = np.zeros((num_hub_cities, num_hub_cities, num_days), dtype=np.int8)
# Set the objective function
objective = 0
for r in range(num_vehicle_types):
    for i in range(num_hub_cities):
        for j in range(num_hub_cities):
            if(i != j):
```

```python
# Add the ad hoc route cost constraint
ad_hoc_route_cost_constraint = solver.Constraint(0, 0.3)
for i in range(num_hub_cities):
    for j in range(num_hub_cities):
        for t in range(num_days):
            ad_hoc_route_cost_constraint.Add(y[i, j, t] * df['variable_cost'][3] >= 0)

# Solve the problem
solver.Solve()

# Get the solution
routes = []
for t in range(num_days):
    route = []
    for r in range(num_vehicle_types):
        for i in range(num_hub_cities):
            for j in range(num_hub_cities):
                if x[r, i, j, t].solution_value() > 0:
                    route.append((i, j, r))
    routes.append(route)
```

```
# Add the capacity constraints
for r in range(num_vehicle_types):
    for t in range(num_days):
        capacity_constraint = solver.Constraint(0, df['capacity'][r])
        for i in range(num_hub_cities):
            for j in range(num_hub_cities):
                if (i != j):
                    capacity_constraint.Add(x[r, i, j] * df['load_volume'][i, j, t] >= 0)
# Add the due time constraints
for i in range(num_hub_cities):
    for j in range(num_hub_cities):
        for t in range(num_days):
            due_time_constraint = solver.Constraint(0, df['due_time'][i, j, t])
            for r in range(num_vehicle_types):
                due_time_constraint.Add(x[r, i, j] * df['travel_time'][i, j] >= 0)
```

```
# Print the solution
for route in routes:
    print(route)
```

Conclusion

Designing a line haul route network with heterogeneous vehicles to meet the varying origin-destination demand requirements for a period of 15 days is a complex problem. However, by following the solution methodology described above, it is possible to find a solution that is both feasible and cost-effective.

Thank you